Chapter 6
The Measurement Model and Uncertainty



This chapter introduces the basic terms and models used to quantify the uncertainty of both direct and indirect measurements. Basic definitions such as Type A and Type B evaluation of uncertainty are introduced, and the GUM approach to the propagation of uncertainties is explained. Case studies are used to illustrate the approaches to uncertainty analyses for both types of measurements.

6.1 Introduction

Measurements can be categorized as either direct or indirect, depending on how the measurement is taken. The basic definitions that are used throughout the chapter for both types of measurements are given first, followed by the two measurement models and their respective uncertainty analyses. The direct measurement model has received less attention in metrology guidelines, while the indirect measurement model has been emphasized in guidelines such as the JCGM 100 (GUM). We believe that direct measurement models, however, can involve subtle features that warrant separate attention. These subtleties appear in a case study involving direct voltage measurements. A second case study involving measurement of neutron yield is used to illustrate indirect measurements and their associated uncertainty analyses. Much of the terminology and methodologies for indirect measurements parallel the presentation in the GUM.

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6.2 Uncertainty Analysis Framework

This section presents an overlying framework for uncertainty analysis based on the development of the appropriate measurement model and analysis for both direct and indirect measurements. The key terms are defined followed by presentation of the conceptual models and analysis approaches.

Sections 6.3 and 6.4 present and illustrate with case studies the estimation techniques that can be applied to both types of measurements.

6.2.1 Standard Uncertainty

Standard uncertainty is defined as the uncertainty of a measurement expressed in terms of the standard deviation of an uncertainty distribution. If the uncertainty distribution is symmetric and approximately normal, this value can be used in a straightforward way to construct uncertainty intervals with specified confidence levels. If the uncertainty distribution is highly non-normal, the standard deviation may not be as useful, and other measures of dispersion should be used.

6.2.2 Type A Uncertainty Evaluation

The Type A uncertainty method of evaluation is defined in the GUM as the "method of evaluation of uncertainty by the statistical analysis of series of observations." These data typically come from a series of tests performed as part of a planned measurement experiment. Examples include data from Gauge R&R experiments performed on the factory floor, small experiments to confirm the performance of secondary standards, or larger controlled experiments used to evaluate high-precision equipment in a primary standards laboratory. Sources of uncertainty that are not part of a statistical analysis are handled by the Type B evaluation of uncertainty.

6.2.3 Type B Uncertainty Evaluation

The Type B uncertainty method of evaluation is defined in the GUM as the "method of evaluation of uncertainty by means other than the statistical analysis of series of observations." Sources of Type B uncertainties may include historical data, calibration reports, manufacturer's specifications, and reference data from handbooks. Type B uncertainty evaluations attempt to account for variation that is not observable in Type A uncertainty evaluations. For example, long-term tester drift would

typically not affect the uncertainty observed in a short-term experiment. Systematic bias would also not affect the variation observed in a short-term experiment. Type B uncertainty evaluations are thus included as part of the overall uncertainty to give validity to the reported uncertainty interval over an extended period of time. In practice, both Type A and Type B evaluations may involve statistical analysis of series of observations. The distinction is that Type A evaluations are based on immediate time-of-test data, while Type B evaluations are not. The values reported as Type B uncertainties, however, have often been derived from statistical analysis of historical data. As such, caution must be used to ensure that Type B uncertainty evaluations do not "double count" uncertainties present in Type A uncertainty evaluations.

6.2.4 Combined Standard Uncertainty

The combined standard uncertainty is expressed in terms of the standard deviation of an uncertainty distribution obtained by combining the individual variances and covariances of each Type A and Type B evaluation that contribute to the overall uncertainty. The combined standard uncertainty is determined by the root sum of squares (RSS) of those variance and covariance terms.

6.2.5 Confidence Level and Expanded Uncertainty

The conceptual form of a measurement uncertainty interval for either direct or indirect measurements is given by the following:

$$\begin{aligned} &\text{Mean} \pm \text{Measurement Uncertainty, or} \\ &\text{Mean} \pm k\,(\text{Combined Standard Uncertainty}). \end{aligned} \tag{6.1}$$

The measurement uncertainty in this expression is referred to as the expanded uncertainty, defined in more detail below. In terms of the mean of a measurand, coverage factor k, and combined standard uncertainty, this interval can be expressed as

$$\bar{y} \pm t_p(\upsilon_{\text{eff}})u_c(y). \tag{6.2}$$

The remainder of this section will provide additional definitions and show how to calculate the sample mean $\bar{y}$, the multiplier $t_p(\upsilon_{\text{eff}})$, and the combined standard uncertainty $u_c(y)$. In Eq. (6.2) $\bar{y}$ is an average of a Type A series of measurements. For direct measurements it is usually the simple average of a series of measurements, while for indirect measurements it is evaluated using a measurement equation.

$$\begin{aligned} U &= t_p(\upsilon_{\text{eff}}) \cdot u_c(y) \\ &= t_p(\upsilon_{\text{eff}}) \sqrt{u_A^2(y) + u_B^2(y)}. \end{aligned} \tag{6.7}$$

The formulas introduced in Sect. 6.2 apply to both the direct measurement case and the indirect measurement case, although the Type A and Type B standard uncertainties are computed in different ways. The direct measurement problem, the direct measurement model, and the uncertainty analysis of a direct voltage measurement are demonstrated in Sect. 6.3.

For both the direct measurement and indirect measurement uncertainty analyses, care must be taken to distinguish between the uncertainty of a single measurement and the uncertainty of an average of multiple measurements. Both cases are important, depending on the measurement problem under consideration. To accept or reject manufactured parts based on a single measurement, for example, the uncertainty of that single measurement is of interest. To quantify the uncertainty of a device such as a voltage standard, typically multiple measurements of the standard are made and the uncertainty of the average measurement will be of interest.

6.3 Direct Measurements and the Basic Measurement Model

A *direct measurement* is defined as one in which the value of the measurand is read directly from the measuring instrument. Examples include measurements of weight, length, and time, where the value of the measurand is determined with a simple measuring device. In some cases, a direct measurement may be adjusted by environmental or other time-of-test factors, making it an effectively indirect measurement.

In practice, the two types of measurements are not always easily distinguished from one another but will depend on one's point of view. For example, an ohmmeter measures resistance by applying a current, measuring the resulting voltage, and dividing the two. From the perspective of the ohmmeter manufacturer, the resistance measurement is indirect, but from the perspective of the ohmmeter user, the resistance measurement is direct.

The simple conceptual model proposed for direct measurements is as follows, with y representing the measured value of the measurand:

$$y = \text{Actual} + \text{Bias} + \varepsilon_A + \varepsilon_B. \tag{6.8}$$

In this expression, the measured value (y) of the measurand is equal to the actual value plus a potential constant bias term, plus ε_A and ε_B, the Type A and Type B random error terms, respectively.

$$u_B(y) = \frac{U}{2}. \tag{6.13}$$

A more conservative value for the Type B standard uncertainty assumes a Rectangular (Uniform) distribution Eq. (4.6) for the Type B source of uncertainty, resulting in the standard uncertainty:

$$u_B(y) = \frac{U}{\sqrt{3}}. \tag{6.14}$$

This distribution may be appropriate if little is known about the shape of the distribution. In this case, $\frac{U}{\sqrt{3}}$ estimates the standard deviation of the Uniform distribution on the interval $(-U, U)$.

A less conservative value for the Type B standard uncertainty assumes an underlying Triangular distribution Eq. (4.7) for the Type B source of uncertainty, resulting in the standard uncertainty:

$$u_B(y) = \frac{U}{\sqrt{6}}. \tag{6.15}$$

The triangular distribution has been used for resolution uncertainty when reading an analog gauge or ruler, but in practice the Uniform distribution is most often used when no other information is provided because of its conservatism. Fig. 4.12 compares the normal, uniform, and triangular distributions used for this problem.

If the Type B value U is treated as exactly known, then $u_B(y)$ can be treated as exactly known and its associated degrees of freedom can be set to $+\infty$. If this assumption cannot be made, a subjective estimate of the relative uncertainty in $u_B(y)$ can be used to define the Type B degrees of freedom as

$$\upsilon_B \cong \frac{1}{2}\left[\frac{\Delta u_B(y)}{u_B(y)}\right]^{-2}. \tag{6.16}$$

In this expression, the quantity inside the brackets is the relative uncertainty in $u_B(y)$. The rationale for this calculation is discussed in the GUM (G.4.2). The use of this equation will be discussed further in Sect. 11.5.

The methodology recommended for direct measurements is presented next with a case study involving the direct measurement of voltage.

6.3.1 Case Study: Voltage Measurement

A voltage measurement uncertainty study was performed on a National Instruments PXI-5124 digitizer (the tester) using a Tektronix TDS 3024 digitizer as the standard. The purpose of the study was to quantify the uncertainty of the standard digitizer and

such as this would potentially be used to inform decisions regarding product acceptance and product state of health over time.

6.3.2 Discussion

The case study above illustrates the recommended approach for determining the expanded uncertainty associated with direct measurements. A tester digitizer was compared to a standard digitizer for 30 shots of an output voltage source set to 350 V. The standard digitizer uncertainty, tester digitizer uncertainty, and bias uncertainty were determined. The resulting expanded uncertainties were used to derive the tester digitizer's bias-adjusted 95% ($k = 2$) uncertainty interval.

Note that the uncertainty interval above Eq. (6.27) is the uncertainty associated with a single measurement of voltage using the tester digitizer. The uncertainty associated with measuring an average of multiple measurements would use a Type A uncertainty adjusted by the sample size. That is, the Type A uncertainty of the tester digitizer based on sample size n would be

$$u_A(\bar{y}) = \frac{0.90}{\sqrt{n}}. \tag{6.28}$$

The standard digitizer added uncertainty to the tester digitizer uncertainty only through the bias calculation. This is because the Type B uncertainty for the tester digitizer was derived from its tabled calibration result which did not involve the standard digitizer. The standard's Type A uncertainty was larger than the tester's Type A uncertainty because the standard available at the time-of-test was an old standard that had yet to be updated. Despite this limitation, the standard digitizer did provide useful information regarding the bias without significantly inflating the calculated uncertainty of the tester digitizer.

As recommended in both the GUM (3.2.4) and in Phillips and Eberhardt (1997), the uncertainty in the estimate of bias should be included in the expanded uncertainty calculation. Because the estimate of bias was based on 30 paired differences, however, it did not contribute much to the total uncertainty.

6.4 Indirect Measurements and the Indirect Measurement Model

In most measurement studies, the measurand is not measured directly, but is determined through a functional relationship. An *indirect measurement* is a measurement for which the value of the measurand is obtained by measuring other quantities functionally related to the measurand. In the NSE, examples include the

$$\bar{y} = \frac{1}{n}\sum_{k=1}^{n} y_k = \frac{1}{n}\sum_{k=1}^{n} f(x_{1,k}, x_{2,k}, \ldots, x_{N,k}) \tag{6.37}$$

if the measurement equation is highly nonlinear. That is, $\bar{y}$ is taken as the arithmetic mean of the n independent determinations y_k.

6. The next step is to determine the expanded uncertainty, U, which provides an uncertainty interval for the measurand of the form $(\bar{y} - U, \bar{y} + U)$. This interval is expected to contain a large fraction of the values that could reasonably be attributed to the measurand Y. U is determined by multiplying the combined standard uncertainty by a coverage factor $k = t_p(\nu_{eff})$. The effective degrees of freedom ν_{eff} are determined from the W-S approximation Eq. (6.50) with sensitivity coefficients included and $t_p(\nu_{eff})$ is the t-value discussed in Sect. 6.2.
7. This results in the expanded uncertainty:

$$U = ku_c(y). \tag{6.38}$$

8. The final step is to report the uncertainty. The interval should be expressed as

$$\bar{y} \pm U. \tag{6.39}$$

The confidence level p should be reported as well as the value of k used to obtain U.

The methodology presented here for indirect measurements is illustrated in Sect. 6.4.1 with a case study involving the lead probe, a secondary standard used to measure neutron yield.

6.4.1 Case Study: Neutron Yield Measurement

Many neutron measurements made within the NSE are referenced to a lead probe measurement via calibration. The lead probe is a secondary standard, maintained by the Primary Standards Lab (PSL), and is itself referenced to a primary standard lead probe, which is referenced to an ion beam neutron source that produces a known number of neutrons. The uncertainty of any neutron measurement made using a neutron detector is therefore linked to the uncertainty of the lead probe used to calibrate that detector. This uncertainty helps to inform decisions regarding characterization and qualification of various neutron sources.

The case study in this section illustrates the uncertainty analysis of a lead probe measurement of neutron yield following the steps outlined in Sect. 6.4 above for indirect measurements. It is adapted from Walsh et al. (2017). The R code used to perform the calculations is included in the text.

```
$F))
veff
## [1] 28.63258
```

It should be noted that the W-S approximation for the indirect measurement case appears slightly different than for the direct measurement equation Eq. (6.19), as the sensitivity coefficients are included here. The expanded uncertainty is calculated using the coverage factor determined from the t-distribution for a 95% level of confidence (two-sided interval):

$$\begin{aligned} U_{\eta}^{95} &= t_{0.975}(29) \times u_c(\eta) \\ &= (2.05) \times \left(1.796 \times 10^6\right) \\ &= 3.672 \times 10^6 \end{aligned} \tag{6.51}$$

```
# Uncertainty interval
uEta <- qt(p = .975, df = 29) * uC
uEta
## [1] 3672213
```

The uncertainty interval (95% confidence) for a single measurement of total neutron yield is thus:

$$\begin{aligned} \eta \pm U_{\eta}^{95} &= 5.32 \times 10^7 \pm 3.67 \times 10^6 \\ &= 5.32 \times 10^7 \pm 6.9\% \text{ neutrons.} \end{aligned} \tag{6.52}$$

```
interval <- c(n - uEta, n + uEta)
interval
## [1] 49510130 56854555
```

The resulting interval Eq. (6.52) provides a range of values that could reasonably be expected from a single measurement of neutron yield from a laboratory neutron source using the lead probe secondary standard. The uncertainties, not reported here, were evaluated over a wide range of neutron source settings. These results were used, via a traceability chain, to quantify the uncertainty associated with neutron detectors measuring total output of neutron sources. The uncertainties are used both in product acceptance and characterization of neutron source performance.

6.4.2 Discussion

The case study in Sect. 6.3.1 illustrates the recommended approach for determining the expanded uncertainty associated with indirect measurements. Neutron yield was

```
SBoot <- eta(A = A[1], f = f, F = F[1], S = sample, B = B[1])
varS <- var(SBoot)

# Vary B only
sample <- rnorm(m, B[1], B[2])
BBoot <- eta(A = A[1], f = f, F = F[1], S = S[1], B = sample)
varB <- var(BBoot)
# Calculate variance
totalVar <- sum(varA, varF, varS, varB)
varResults <- t(data.frame(f = 0,
 A = varA/totalVar*100,
 F = varF/totalVar*100,
 S = varS/totalVar*100,
 B = varB/totalVar*100))
colnames(varResults) <- c('Percent of Standard Uncertainty')
varResults
## Percent of Standard Uncertainty
## f 0.000
## A 35.09
## F 55.70
## S 9.20
## B 0.012
```

Knowledge of the relative "components of variation" can suggest ways to improve the measurement. Key results from this analysis were that the measurement uncertainty could be reduced by reducing the uncertainty in the attenuation factor A and the calibration factor F. Examination of Eqs. (6.44) and (6.47), however, shows that significant reduction in uncertainty can be realized by increasing the net scalar counts per source neutron (i.e., reducing f). However, this would require a new detector technology with improved efficiency, requiring substantial design effort.

6.5 Related Reading

The direct measurement model (Eq. 6.8) presented here is typically not covered in metrology guidelines such as the JCGM:100 (GUM). This may be because direct measurements are less common in practice, and their uncertainty analyses often result in intervals of the form

$$\overline{y} \pm k u_c(y),$$

where $\overline{y}$ and $u_c(y)$ are based on simple summary statistics. The model was included in Chap. 6 primarily to introduce the idea of an additive statistical model that could be extended to include multiple fixed and random terms in a more complicated ANOVA (Sect. 9.3). A metrology guideline that does cover the direct measurement model is ISO 5725-1: 1994, which provides guidelines in designing, implementing,

approximation may not work well when there is a significant difference in the magnitude of uncertainty terms. In particular, adding small uncertainty terms with large degrees of freedom to a much larger term with small degrees of freedom results in a distribution not well approximated by the W-S approximation.

The Monte Carlo method outlined in the GUM Supplement 1 will often be an appropriate alternative in these special cases. That is the subject of Chap. 8.

6.6 Exercises

6.1. Scientists in Hawaii want to assess the uncertainty of a direct thermocouple measurement of lava temperature when it first flows from a volcano. A sample of ten measurements appears in Table 6.6. Compute the Type A standard uncertainty for this measurement.

6.2. A calibration report for the thermocouple under investigation reports a tolerance of $\pm$ 75 °C over the range 1000–1300 °C. Compute the Type B standard uncertainty associated with the calibration of this thermocouple. Assume a uniform distribution (4.4.1.1) over the stated tolerance interval.

6.3. Compute the combined standard uncertainty and the effective degrees of freedom using the Welch–Satterthwaite approximation. What assumption would you make regarding Type B degrees of freedom?

6.4. Compute the coverage factor for a confidence level of 95.45%. Report the result of the measurement along with the expanded uncertainty.

6.5. How could you improve the measurement of volcanic lava temperature?

6.6. Suppose an indirect measurement is defined by the equation $Y = f(X_1, X_2) = X_1^2 + X_2^2$, where X_1 and X_2 are independent with $N(0, 1)$ distributions. Using a first-order approximation, estimate the expanded uncertainty of the measurand Y. Is this a good approximation? Why or why not?

Table 6.6 Lava temperature data

Reading number	Temperature (°C)
1	1130.60
2	1210.64
3	1211.25
4	1047.01
5	1135.67
6	1189.19
7	1237.59
8	1164.33
9	1225.51
10	1165.79
$\bar{y}$	1171.76

Table 6.7 Mean input quantities and standard uncertainties for XRF measurement

Input quantity	Mean (μ)	Standard uncertainty Type A	Type A degrees of freedom	Type A PDF
$Y_{\text{uncorrected}}$	0.4409 μm	0.00360	19	Normal
X_1	0.1820 μm	0.00093	9	Normal
X_2	0.1823 μm	0.00058	19	Normal

6.7. Now assume X_1 and X_2 are independent with $N(1000, 1)$ distributions. Using a first-order approximation, estimate the expanded uncertainty of the measurand Y. Is this a good approximation? Why or why not?

6.8. What steps should be followed in developing an indirect measurement equation? What is a "hybrid" measurement equation? When might you use one?

6.9. A corrected X-ray fluorescence (XRF) thickness measurement, $Y_{\text{corrected}}$, is estimated indirectly from an uncorrected thickness measurement, $Y_{\text{uncorrected}}$, via the relationship

$$Y_{\text{corrected}} = \left(\frac{X_1}{X_2}\right) \cdot Y_{\text{uncorrected}},$$

where X_1 and X_2 are independent correction factors. Table 6.7 above provides the information needed to complete an uncertainty analysis. The Type B uncertainties were negligible and were not part of the analysis.
Using the GUM methodology (Sect. 6.4), perform an uncertainty analysis for the measurand $Y_{\text{corrected}}$.

6.10. How is a "components of variation" analysis used as part of an uncertainty analysis? Perform this analysis for the measurand $Y_{\text{corrected}}$.

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